



# SAJAG AI

सजग = alert, vigilant

The on-device, multimodal scam shield for Snapdragon AI PCs

*Sees the scam on your screen. Hears it on your call. Stops it before money moves. Nothing leaves your PC.*

## 5 AI models

on the Hexagon NPU path

EN | हिन्दी | Hinglish

speech + screen

## 100% offline

zero cloud, zero upload

91% / 0.5%

detection / false alarms\*

**Tanmay Sahare**

B.Tech Aerospace Engineering, IIT Kanpur | [github.com/tanmaysahare/sajag-ai](https://github.com/tanmaysahare/sajag-ai)

\*synthetic evaluation, see Results



Built for Snapdragon-powered HP AI PCs

# Scams now happen live, on the PC screen and the call

Every defence today acts before or after the call. Nothing guards the moment the scam happens.

₹

₹22,845 cr

lost to cyber fraud in 2024, up 206% vs 2023 (MHA)

💣

1,23,672

digital-arrest cases in 2024, ₹1,918 cr lost (I4C)

👤

7 in 10

Indians met a tech-support scam in a year, highest globally (Microsoft)

🎭

47%

of Indians faced or know an AI voice-clone scam (McAfee)

Digital-arrest scams: cases fell, loss per case more than doubled

| Year                     | 2022 | 2023 | 2024  | 2025 |
|--------------------------|------|------|-------|------|
| Loss (₹ crore)           | 91   | 339  | 1,918 | 644  |
| Avg loss / case (₹ lakh) | 2.28 | 5.59 | 1.55  | 3.73 |

Awareness drives cut 2025 cases by 86%, but each remaining victim lost 2.4x more than in 2024. Warnings work; they must arrive at the moment of the call.

📞

Contact

WhatsApp / Skype video call, or a full-screen browser pop-up

🏛️

Authority

'CBI', 'Mumbai Police', 'Microsoft support', fake warrant on screen

⚠️

Fear

Drugs in your parcel, Aadhaar misused, PC infected, arrest warrant

🔒

Isolation

Stay on camera, do not tell anyone, do not disconnect

💻

Control

Install AnyDesk / TeamViewer, share screen

⚡

Money

Transfer to 'RBI safe account', read the OTP

THE KILL CHAIN

Why existing defences miss it

| Defence                          | Before | During | After  |
|----------------------------------|--------|--------|--------|
| Awareness ads, caller tunes      | ✓      | ✗      | ✗      |
| Bank SMS: 'never share OTP'      | ✓      | ✗      | ✗      |
| Spam caller-ID apps              | Partly | ✗      | ✗      |
| Browser scareware blockers       | ✗      | Partly | ✗      |
| 1930 helpline, cybercrime.gov.in | ✗      | ✗      | ✓      |
| Sajag AI (on-device)             | ✓      | ✓      | Partly |

Only on-device AI can watch the screen and the call without uploading them. A cloud version would itself be a privacy risk.

Problem statement:

Indians are being defrauded live on their own PCs by scammers who impersonate police, banks and tech support in Hindi, English and Hinglish. The victim sees and hears every signal, but under fear and isolation cannot act on it. We need a guardian that understands the call and the screen together, in Indian languages, in real time, and without the data ever leaving the device.

15.9 M PCs

shipped in India in 2025, a record; HP leads the market (IDC)

# Who gets scammed on a PC, and what each of them needs

Personas built from MHA, I4C, RBI advisories and reported cases

**Retired parent, 60+**  
Pensioner, Nagpur. Video-calls children on the laptop, uses net banking.

**Pain points**

- Targeted by 'digital arrest' and fake CBI calls
- Fear of authority, kept on camera for hours
- Senior citizens lost ₹ crores city by city in 2025

**Typically hears:** 'Aapke naam se arrest warrant hai, camera on rakho'

**What Sajag does:** Full-screen Hindi warning + voice, tells them to call family / 1930

**Young professional, 22-35**  
WFH laptop, Bengaluru. Heavy WhatsApp Desktop and Teams user.

**Pain points**

- 'FedEx parcel with drugs' and task-job scams
- Hinglish calls from 'customs' or 'HR'
- Trusts screen-share requests at work

**Typically hears:** 'Your FedEx parcel has MDMA, connecting you to Mumbai police'

**What Sajag does:** Hinglish call understanding + remote-access alert

**Small-business owner, 35-55**  
Shop / CA office, Kanpur. Banking and GST portals open all day.

**Pain points**

- KYC-expiry and 'refund' calls with OTP asks
- 'Support' installs AnyDesk on the billing PC
- Money screens open while on calls

**Typically hears:** 'Sir, KYC expired. Share the OTP or account will be blocked'

**What Sajag does:** Fusion: remote access + payment screen + OTP ask = DANGER

**College student, 18-24**  
Hostel laptop. Telegram, YouTube, trading apps.

**Pain points**

- 'Like videos, earn ₹5,000 a day' task scams
- Fake IPO / crypto groups with guaranteed returns
- Pay-to-withdraw 'earnings'

**Typically hears:** 'Pay ₹4,999 tax to withdraw your ₹48,000 profit'

**What Sajag does:** Advance-fee rule: prize or profit that must be paid for

| Scam families Sajag knows (7) x tactics (10) |                                     |                                           |
|----------------------------------------------|-------------------------------------|-------------------------------------------|
| Family                                       | Signature tactics                   | Example cue (as spoken)                   |
| Digital arrest                               | authority, threat, secrecy, payment | 'kisi ko mat batana, call mat kaatna'     |
| Tech support                                 | fake_error, remote_access, payment  | 'download AnyDesk, tell me the code'      |
| KYC / bank                                   | identity_bait, threat, credential   | 'केवाईसी खत्म, ओटीपी बताइए'               |
| UPI refund                                   | lure, credential                    | 'enter UPI PIN to receive refund'         |
| SIM / electricity                            | authority, threat, urgency          | 'sim band ho jayega 2 ghante mein'        |
| Investment / task job                        | lure, payment                       | 'pakka munafa, pehle task ke paise bhejo' |

**Interrupt, don't whisper**  
victims are in a fear state; small toasts get ignored

**Speak their language**  
Hindi, English and Hinglish in both speech and screen

**Explain the why**  
show the exact phrase and source behind every alert

**Never phone home**  
no account, no cloud, works offline



# 15 candidate solutions, scored against the judging criteria

Weights from an AHP pairwise comparison (next slide). Scores 1-5.

| #  | Candidate solution for Snapdragon HP AI PCs                             | Tech depth on NPU (w=0.31) | Innovation (w=0.19) | India impact (w=0.17) | Deploy & access (w=0.14) | Why on-device (w=0.09) | Buildability (w=0.10) | Weighted    |
|----|-------------------------------------------------------------------------|----------------------------|---------------------|-----------------------|--------------------------|------------------------|-----------------------|-------------|
| 1  | <b>Sajag: on-device multimodal scam shield (screen + call + system)</b> | 5                          | 5                   | 5                     | 4                        | 5                      | 4                     | <b>4.76</b> |
| 2  | Deepfake video-call detector (face + voice liveness)                    | 5                          | 5                   | 4                     | 3                        | 4                      | 1                     | <b>4.06</b> |
| 3  | Offline crop-disease + Hindi voice advisor for CSC / FPO kiosks         | 4                          | 3                   | 5                     | 3                        | 4                      | 3                     | <b>3.74</b> |
| 4  | Drone-video triage ground station for disaster response                 | 5                          | 4                   | 3                     | 2                        | 4                      | 2                     | <b>3.67</b> |
| 5  | Rural clinic medical scribe (ASR + note LLM)                            | 4                          | 3                   | 4                     | 3                        | 4                      | 3                     | <b>3.57</b> |
| 6  | Indian Sign Language to speech translator                               | 4                          | 4                   | 4                     | 3                        | 3                      | 2                     | <b>3.57</b> |
| 7  | DPDP-compliance PII redaction shield for SMEs                           | 3                          | 3                   | 3                     | 4                        | 5                      | 5                     | <b>3.52</b> |
| 8  | Offline Indic KYC / form assistant for CSC operators                    | 3                          | 3                   | 3                     | 4                        | 4                      | 4                     | <b>3.33</b> |
| 9  | Low-vision scene describer (VLM on NPU)                                 | 4                          | 2                   | 4                     | 3                        | 3                      | 3                     | <b>3.29</b> |
| 10 | Energy-aware AI workload router (CPU / GPU / NPU)                       | 5                          | 3                   | 1                     | 3                        | 3                      | 3                     | <b>3.28</b> |
| 11 | Always-on posture & eye-strain coach                                    | 3                          | 2                   | 2                     | 5                        | 4                      | 5                     | <b>3.20</b> |
| 12 | Offline Indic live captions for classrooms                              | 3                          | 2                   | 3                     | 4                        | 4                      | 4                     | <b>3.13</b> |
| 13 | Local RAG second brain over personal files                              | 3                          | 1                   | 2                     | 5                        | 4                      | 5                     | <b>3.01</b> |
| 14 | Private meeting / lecture copilot                                       | 3                          | 1                   | 2                     | 5                        | 3                      | 5                     | <b>2.92</b> |
| 15 | On-device AI exam proctor                                               | 3                          | 2                   | 2                     | 3                        | 4                      | 4                     | <b>2.83</b> |

## How ideas were generated

- Mapped AI Hub's model catalogue (VAD, Whisper, OCR, LLMs, VLMS, pose, detection) to Indian problems where on-device matters
- Scanned 10+ public challenge repos to avoid crowded themes (study buddies, note-takers, health bots)
- Filtered by what a judge can verify: runnable demo, measurable NPU path
- Scored each idea on 6 criteria mapped to the 4 official judging criteria

## Why the runner-ups lost

**Deepfake call detector:** novel but no validated Indian dataset; risky claims

**Crop-disease kiosk:** farmers use phones, not PCs; lab datasets fail in field

**Drone triage:** needs aerial fine-tuning and hardware to demo

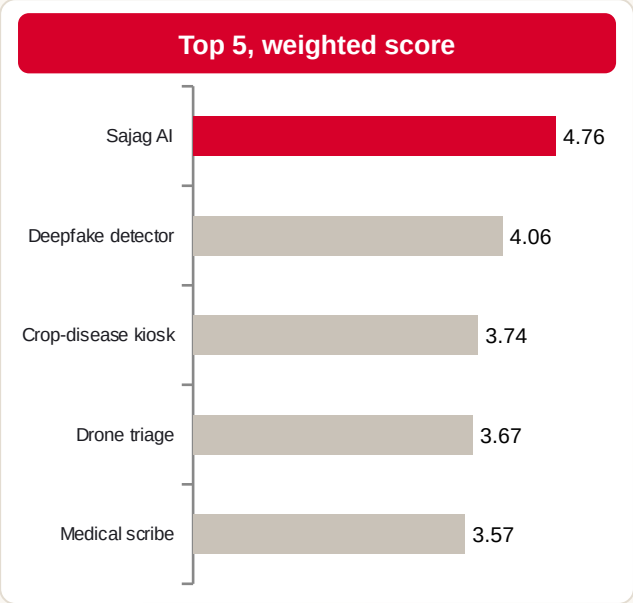
**Meeting copilot / RAG:** crowded, already built into Windows and AnythingLLM

# Why Sajag: AHP-weighted choice and a clear gap in the market

Consistency ratio 0.013 (< 0.10 means the pairwise judgements are consistent)

| AHP pairwise comparison of criteria |     |     |     |     |    |    |        |
|-------------------------------------|-----|-----|-----|-----|----|----|--------|
| Criteria                            | C1  | C2  | C3  | C4  | C5 | C6 | Weight |
| C1 Technical                        | 1   | 2   | 2   | 2   | 3  | 3  | 0.311  |
| C2 Innovation                       | 1/2 | 1   | 1   | 2   | 2  | 2  | 0.192  |
| C3 India                            | 1/2 | 1   | 1   | 1   | 2  | 2  | 0.171  |
| C4 Deployment                       | 1/2 | 1/2 | 1   | 1   | 2  | 1  | 0.136  |
| C5 Why                              | 1/3 | 1/2 | 1/2 | 1/2 | 1  | 1  | 0.090  |
| C6 Buildability                     | 1/3 | 1/2 | 1/2 | 1   | 1  | 1  | 0.101  |

C1 Technical depth on NPU, C2 Innovation, C3 India impact, C4 Deployment & accessibility, C5 Why on-device, C6 Buildability & demo. C1 carries the most weight because Technical Implementation is the first judging criterion and the official tie-breaker.



- What makes Sajag different
- Multimodal:** call + screen + system fused over time, not one message at a time
  - Indian languages:** Hindi, English, Hinglish cues in Devanagari and Latin script
  - PC-native:** works with any call app via audio loopback, no integration needed
  - Private by construction:** nothing stored, nothing uploaded
  - Explainable:** every alert cites the phrase and the source
  - Measurable:** 27 tests, reproducible evaluation, on-device bench command
  - Portable:** the same AI Hub models run on Snapdragon phones

| Benchmark against what exists today (public documentation, Sept 2026) |          |                        |                                 |                       |                      |                       |
|-----------------------------------------------------------------------|----------|------------------------|---------------------------------|-----------------------|----------------------|-----------------------|
| Capability                                                            | Sajag AI | Edge Scareware blocker | Google on-device scam detection | Caller-ID / spam apps | Bank SMS & awareness | Sanchar Saathi / 1930 |
| Acts during the scam, in real time                                    | ✓        | ✓                      | ✓                               | Partly                | ✗                    | ✗                     |
| Understands call audio (Hindi / Hinglish)                             | ✓        | ✗                      | Partly                          | ✗                     | ✗                    | ✗                     |
| Reads any screen / app, not only the browser                          | ✓        | ✗                      | ✗                               | ✗                     | ✗                    | ✗                     |
| Detects remote-access tools during a call                             | ✓        | ✗                      | ✗                               | ✗                     | ✗                    | ✗                     |
| Runs fully on-device, no upload                                       | ✓        | ✓                      | ✓                               | ✗                     | n/a                  | n/a                   |
| Works on Windows PCs (WhatsApp, Teams, Skype, Zoom)                   | ✓        | Partly                 | Partly                          | ✗                     | Partly               | Partly                |
| Explains why, in the user's language                                  | ✓        | Partly                 | Partly                          | ✗                     | Partly               | ✗                     |



# Sajag in action: a digital-arrest call, second by second

Replayed by `python -m sajag demo digital\_arrest` (real OCR, real risk engine)

Timeline

t=0s

WhatsApp video call starts. System sentinel marks a live call.

SAFE

t=4s

'Main Mumbai police cyber crime se Inspector bol raha hoon' (Whisper, Hinglish)

SAFE 13%

t=12s

'Aapke naam se FedEx parcel mein drugs mile hain'

CAUTION 46%

t=20s

'Aadhaar used in money laundering, arrest warrant issued'

WARNING 61%

t=26s

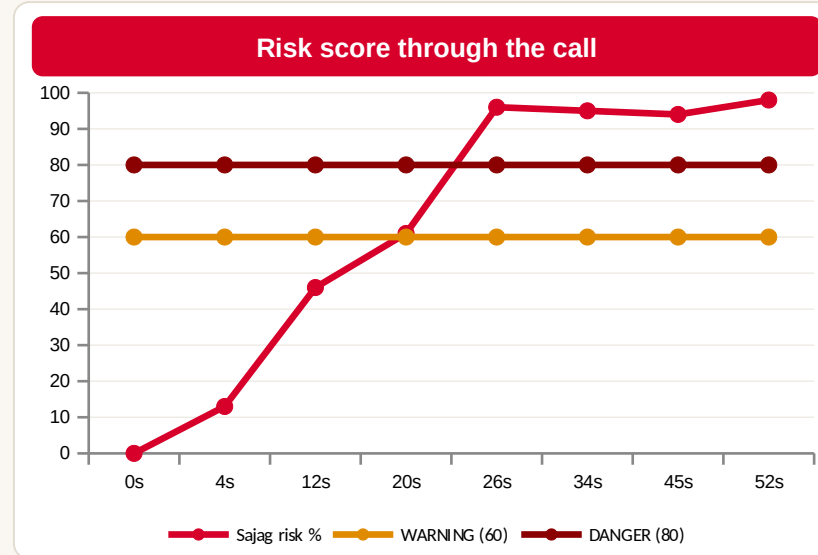
Fake CBI warrant appears on screen. NPU OCR reads 'digital arrest', 'do not tell anyone'

DANGER 96%

t=45s

'Transfer to RBI safe account'. Sajag has already interrupted, 19 s earlier.

DANGER 94%



The interrupt (DANGER)

STOP. This looks like a scam.

रुकिए! यह धोखाधड़ी लगती है।

Digital arrest / fake police

No police, CBI or court arrests anyone over a video call or asks you to move money for 'verification'.

Hang up. Tell your family. Call 1930.

I understand, close

Full-screen, bilingual, spoken aloud. Lists the reasons and the 1930 helpline.

Live dashboard (127.0.0.1, offline)

Sajag AI

On-device scan shield for Snapdragon AI PCs. Nothing leaves this PC.

CPU fallback (x86\_64)

LIVE RISK

98% DANGER

Digital arrest / fake police

- Threatens arrest, penalty or blocking ('digital arrest', screen)
- Demands secrecy or 'isolation' ('kisi ko mat batana', call)
- Asks for OTP, PIN or card details ('otp', screen)
- Asks you to move money ('payment screen open during a call', system)
- Claims your Aadhaar / SIM / parcel was misused ('aadhaar card has been used', call)
- Impersonates an authority ('fedex', call)

No police, CBI or court ever arrests anyone over a video call or asks you to move money for 'verification'. Hang up, tell a family member, and call 1930 or report at cybercrime.gov.in.

दुर्गति, धोखाधड़ी या अदभुत कहानी कहेंगे आपको हमारे पास नहीं आता। हमारे पास सिर्फ एक ही रास्ता है। हमें 1930 पर कॉल करें या cybercrime.gov.in पर रिपोर्ट करें।

WHAT SAJAG SEES (LIVE)

SCREEN (OCR)

NetBanking Fund Transfer AddPayee/ Beneficiary/RBI SafeAccount Transfermode (MPS/ RTGS AmounttoBepaid: Rs9,80,000 Enter theOneTimePassword(OTP) sent to your mobile SIMULATEDSCAMSCREENFORSAJAGADEMO-NOTAREALDOCUMENT (payment: rgtg, cybercrime.gov))

CALL AUDIO (CALLER)

Verification ke liye apne account ka paisa RBI safe account mein transfer karo, jaanch ke baad wapas milega. (payment: safe account)

CALL AUDIO (CALLER)

We national secret hai, kisi ko mat batana aur call mat kaatna, camera on rakho. (secrecy: kisi ko mat batana)

ALERT

DANGER: Digital arrest / fake police (96%)

SCREEN (OCR)

CENTRALBUREAUOFINVESTIGATION ARRESTWARRANT Case: Money Laundering and Narcotics The accused is under digital arrest until investigation is complete. Keep this confidential. Do not tell anyone. Transfer funds to the RBI's (threat: digital arrest, secrecy: keep this confidential, payment: safe account)

ALERT

DANGER: Digital arrest / fake police (61%)

TRY A SCENARIO

Digital arrest over a WhatsApp video call (Hinglish)

Fake Microsoft pop-up + AnyDesk remote access

KYC expiry call asking for OTP (Hindi)

Normal family video call (should stay safe)

Office IT fixing a laptop over TeamViewer (legit, low risk)

Reset

ON-DEVICE AI

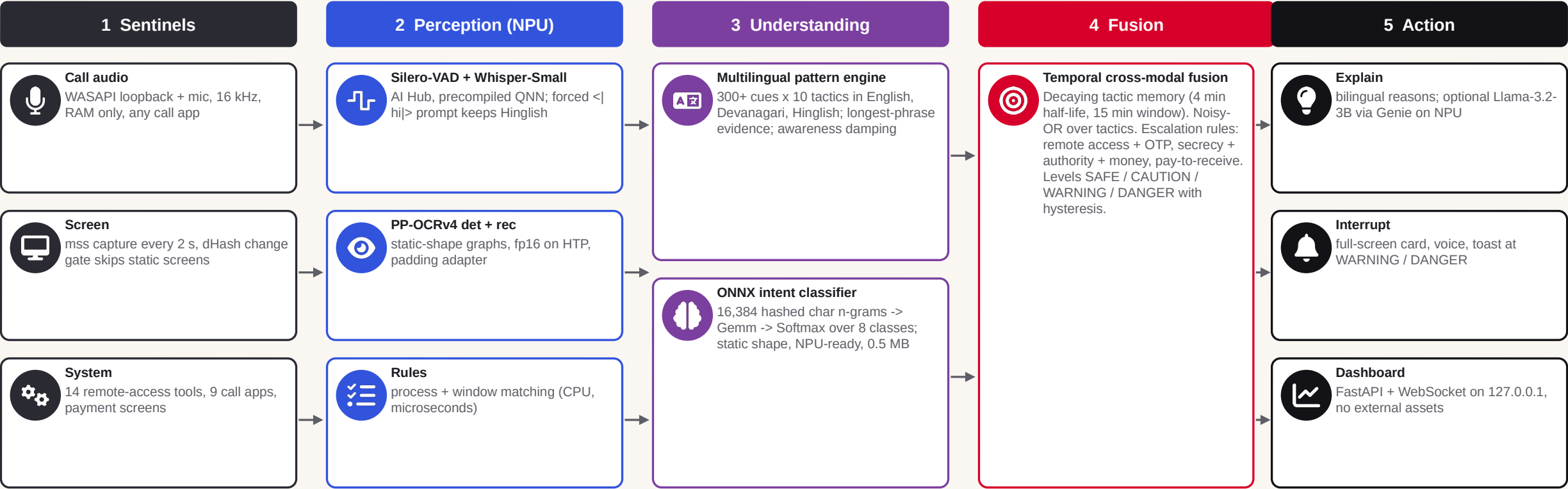
| Model             | Runs on | ms    |
|-------------------|---------|-------|
| Intent-classifier | CPU     | 0.284 |

Segments 5 | screens 2 | skipped 0 | alerts 3

- Risk gauge with family, reasons and advice in English and Hindi
- Live feed of what each sentinel saw, with the matched cue highlighted
- Which model ran on which unit (QNN NPU or CPU) and its latency
- One-click replay of 5 scenarios for judges without a Snapdragon PC

# Architecture: three sentinels, one fusion brain, all on the device

Heterogeneous compute: each stage runs where it is cheapest



| Heterogeneous compute map |           |                  |                          |
|---------------------------|-----------|------------------|--------------------------|
| Stage                     | Unit      | HTP power mode   | Runs when                |
| Silero-VAD                | NPU       | low_power_saver  | call active, every 32 ms |
| PP-OCRv4 det + rec        | NPU       | high_performance | screen changed           |
| Whisper-Small enc + dec   | NPU       | burst            | VAD speech segment       |
| Intent classifier         | NPU / CPU | power_saver      | each segment             |
| Capture, rules, fusion    | CPU       | -                | always (microseconds)    |

Built with

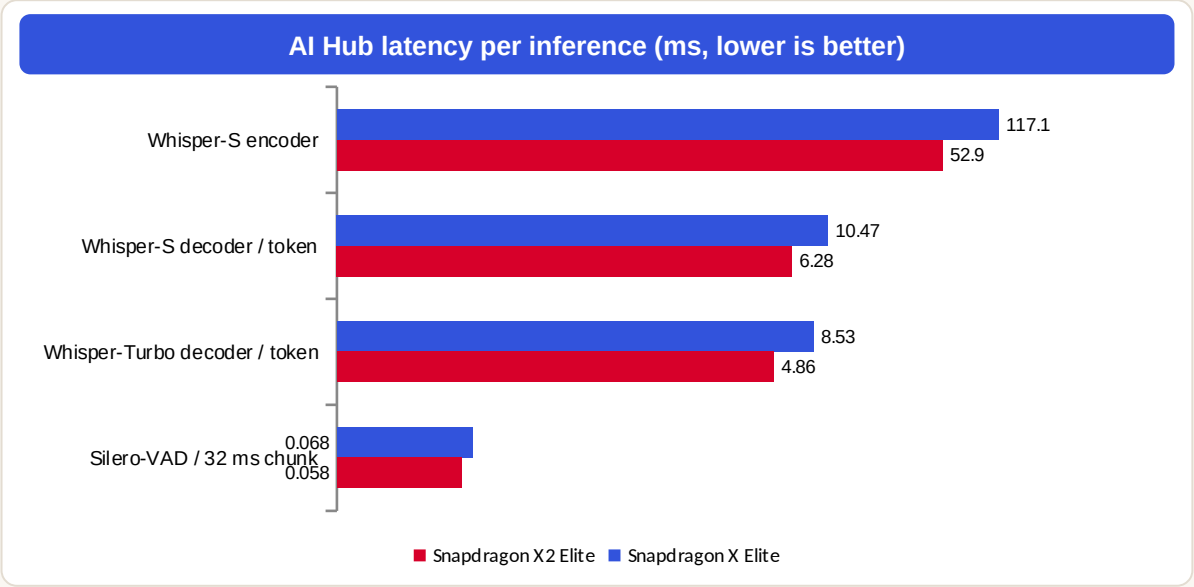


Qualcomm AI Hub | ONNX Runtime QNN EP | Python | PaddleOCR | Llama 3.2 | FastAPI | GitHub | Windows on Arm



# Technical implementation: engineered for the Hexagon NPU

Published Qualcomm AI Hub profiles on Snapdragon X Elite / X2 Elite CRD



| Model stack    |                              |              |                                |
|----------------|------------------------------|--------------|--------------------------------|
| Role           | Model                        | Source       | NPU evidence                   |
| Voice gate     | Silero-VAD (w8a16)           | AI Hub       | 51/51 layers NPU, 0.068 ms     |
| Speech to text | Whisper-Small (multilingual) | AI Hub       | 1582 + 2277 layers, 100% NPU   |
| Speech (X2)    | Whisper-Large-v3-Turbo       | AI Hub       | enc 251 ms, dec 4.9 ms/tok     |
| Screen text    | PP-OCrv4 det + rec           | PaddleOCR    | static shapes, fp16 on HTP     |
| Scam intent    | Sajag classifier             | trained here | Gemm + Softmax, static         |
| Explanation    | Llama-3.2-3B (w4a16)         | AI Hub       | 11.3 tok/s X Elite, 42.5 on X2 |

Six NPU techniques implemented in sajag/npu.py and vision/ocr.py



**Static-shape adapter**  
pads variable OCR inputs into the compiled [1,3,736,1280] canvas, crops outputs, CPU only for oversize lines



**Per-workload power modes**  
low\_power\_saver for sentinels, burst only while a call is live



**EP context caching**  
HTP graph compiled once, context binary reloaded on every launch



**Strict NPU mode**  
SAJAG\_NPU\_STRICT=1 disables silent CPU fallback during development



**Compute gating**  
dHash screen gate + call gate + VAD gate keep the NPU idle most of the time



**Torch-free Whisper**  
AI Hub decode loop ported to NumPy with a 200-slot static KV cache

From code: one factory, any device

```
providers = [{"QNNExecutionProvider", {  
  "backend_path": "QnnHtp.dll",  
  "htp_performance_mode": POWER[workload],  
  "htp_graph_finalization_optimization_mode": "3",  
  "enable_htp_fp16_precision": "1"}},  
  "CPUExecutionProvider"]  
so.add_session_config_entry("ep.context_enable", "1")
```

Same code runs on the NPU on Snapdragon and on the CPU anywhere else, so judges can run the demo on any laptop.

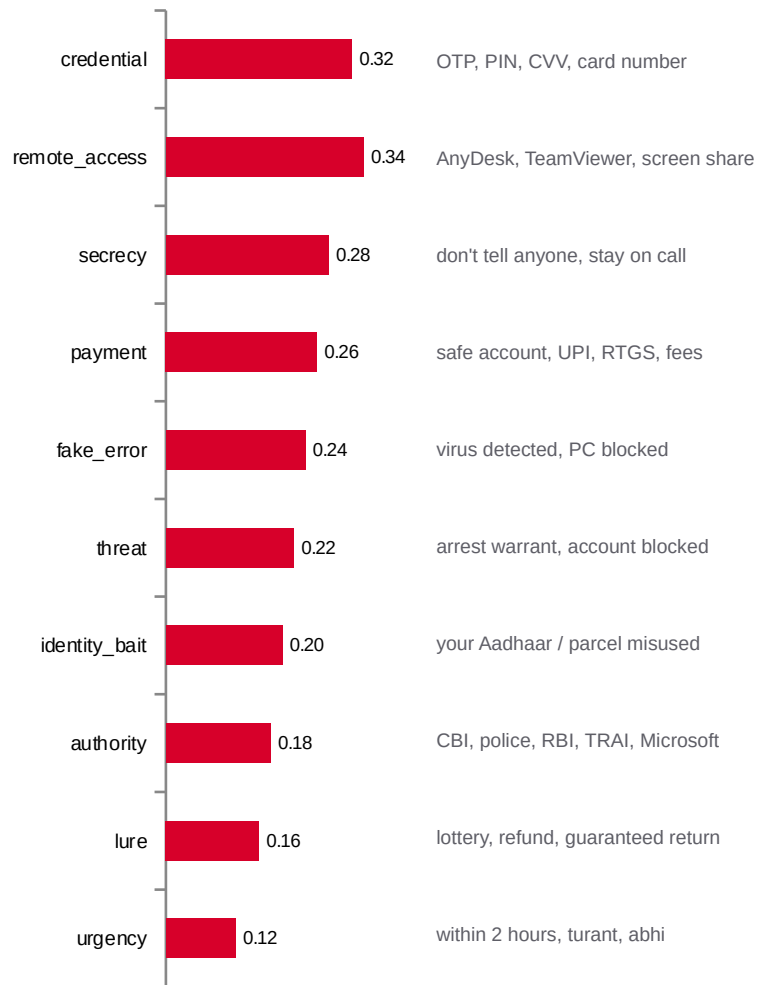




# The risk engine: from a sentence to a verdict you can trust

Deterministic, explainable, tunable. The LLM never decides the verdict.

## 10 tactics, weighted



## Per sentence (segment score)

- Normalise: NFKC, case, punctuation, ASR / OCR confusions ('any desk' to 'anydesk')
- Match 300+ cues; Latin on word boundaries, Devanagari as substrings; keep the longest phrase
- $pattern = 1 - PRODUCT(1 - w\_tactic)$
- $segment = 1 - (1 - pattern)(1 - 0.6 \times p\_scam)$
- Awareness damping x0.35 for 'police never ask...', bank SMS footers

## Across the conversation (fusion)

- Evidence decays with a 4-minute half-life
- Evidence weight scaled by classifier confidence; OTP, remote-access and secrecy asks keep 60% during a live call
- Noisy-OR over the strongest evidence per tactic
- +0.08 when 2+ modalities agree during a call
- Alert fires once per upward level change

## Escalation rules encode what investigators describe

| Signature              | Condition (decayed evidence)                                                | Floor        | Catches                            |
|------------------------|-----------------------------------------------------------------------------|--------------|------------------------------------|
| Remote control + money | $remote\_access \geq 0.20$ and $(payment \text{ or } credential) \geq 0.15$ | 0.85 DANGER  | tech support, KYC with AnyDesk     |
| Digital arrest         | $secrecy \geq 0.12$ , $payment \geq 0.12$ , $authority \geq 0.08$           | 0.90 DANGER  | fake CBI / police / customs        |
| Advance fee            | $lure \geq 0.08$ and $payment \geq 0.12$                                    | 0.68 WARNING | lottery, task job, IPO, refund     |
| Cross-modal agreement  | call live, 2+ sources, $score \geq 0.5$                                     | +0.08        | screen confirms what the call says |

**Worked example:** 'main CBI se bol raha hoon' (authority) + 'kisi ko mat batana' (secrecy) + 'RBI safe account mein transfer karo' (payment) = digital-arrest signature, floor 0.90, DANGER. Any one of those lines alone stays below WARNING.

Thresholds tuned on synthetic calls: false alarms fell from 22% to 0.5% while detection rose from 65.5% to 91% (see Results).

# Results: catches scams early, stays quiet on normal calls

Reproducible with ``python scripts/evaluate.py`` and ``python -m pytest``

✓

91%

of 200 synthetic scam calls reach WARNING

✓

0.5%

false alarms on 200 benign calls with hard negatives

✓

2 turns

median to first WARNING

✓

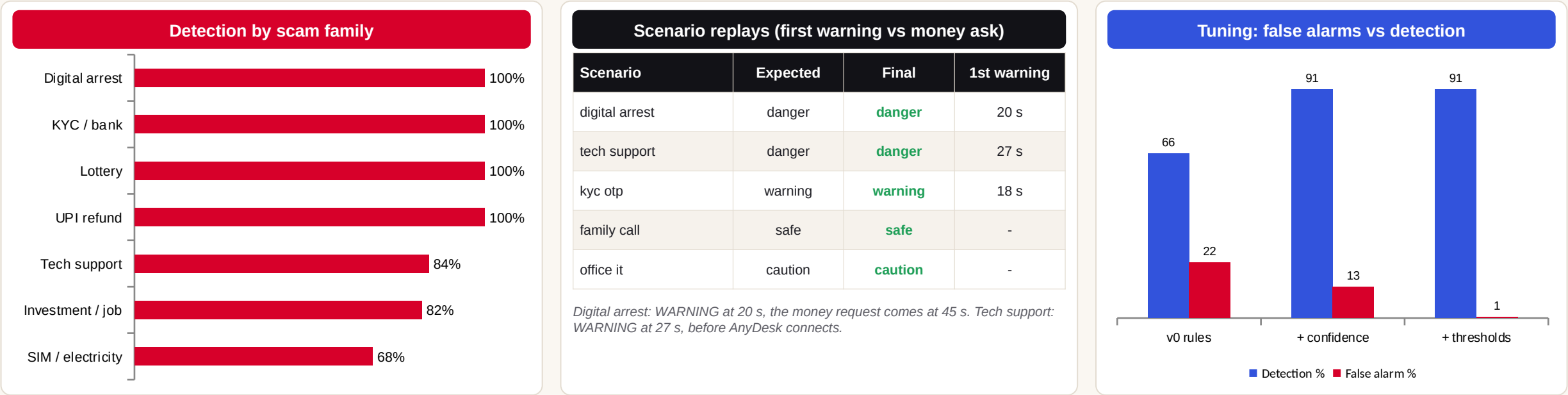
5 / 5

end-to-end scenarios with real OCR pass

✓

27 / 27

unit + pipeline tests pass



Measured latency here (x86 laptop CPU, fallback path)

| Component                                  | p50     | Note                       |
|--------------------------------------------|---------|----------------------------|
| Intent classifier (ONNX)                   | 0.03 ms | static [1,16384] Gemm      |
| Full text analysis (patterns + classifier) | 1.8 ms  | per sentence               |
| Screen OCR det + rec (CPU)                 | ~730 ms | moves to NPU on Snapdragon |

Honest limitations

- Corpus is synthetic and overlaps training templates: numbers are an upper bound
- Classifier alone on held-out templates: precision 1.00, recall 0.40, which is why rules + fusion exist
- NPU latencies quoted from AI Hub; ``python -m sajag bench`` measures them on the device

# Deployment & accessibility: one command, works for anyone

Open source (Apache-2.0), offline, bilingual, elderly-first UX

Install and run on a Snapdragon HP PC

```
git clone github.com/tanmaysahare/sajag-ai
powershell -File install.ps1
python -m sajag doctor # QNN NPU found
python -m sajag run    # guardian + dashboard
```

- install.ps1: Python 3.11 ARM64 venv, onnxruntime-qnn, AI Hub models via qai\_hub\_models\_cli, NPU OCR prep
- Judges on any laptop: `python -m sajag demo` or `python -m sajag ui`

Accessibility by design

**Two languages, three scripts**  
English, हिन्दी, Hinglish in speech, screen and alerts

**Elderly-first alerts**  
full-screen, large type, spoken aloud, one button

**Zero setup after install**  
no account, no sign-in, no subscription

**Works offline**  
no internet needed after the one-time model download

On-device vs a cloud scam detector

|                              | Sajag (NPU) | Cloud AI           |
|------------------------------|-------------|--------------------|
| Screen + call data leaves PC | never       | always             |
| Latency to warning           | sub-second  | network bound      |
| Works offline                | ✓           | ✗                  |
| Cost per user per month      | ₹0          | ASR + LLM API fees |
| DPDP Act data minimisation   | ✓           | Partly             |

Go-to-market: who puts Sajag on PCs

**OEM preload**  
HP leads India's PC market; a Snapdragon HP AI PC could ship with Sajag as a 'Copilot+ safety' feature

**Banks & NBFCs**  
offer to retail and senior customers as part of fraud-prevention programmes

**I4C / 1930 & police**  
awareness partners distribute a free, offline tool; 1930 is built into every alert

**Communities**  
senior-citizen associations, colleges, CSCs and CA offices

Impact model (illustrative, from 2024-25 I4C figures)

| Cases        | Impact (₹ lakh) |
|--------------|-----------------|
| 1,000 cases  | 15.5            |
| 5,000 cases  | 77.5            |
| 10,000 cases | 155.0           |
| 25,000 cases | 387.5           |

**₹1.55 lakh**  
average loss per digital-arrest case in 2024 (₹1,918 cr / 1,23,672 cases); ₹3.73 lakh in 2025.  
Every 1,000 digital-arrest attempts Sajag interrupts protects about ₹15.5 crore at 2024 averages, before counting tech-support, KYC and investment fraud.

# Roadmap: RICE-prioritised, with risks named up front

RICE = Reach x Impact x Confidence / Effort

| Next features, ranked by RICE                          |         |        |       |             |      |
|--------------------------------------------------------|---------|--------|-------|-------------|------|
| Feature                                                | Reach % | Impact | Conf. | Effort (wk) | RICE |
| Real-call validation with bank / I4C partners          | 90      | 3      | 80%   | 2           | 108  |
| Marathi, Tamil, Telugu, Bengali cue libraries          | 70      | 2      | 80%   | 1.5         | 74.7 |
| Family guardian: consented alert to a trusted contact  | 50      | 2      | 80%   | 1.5         | 53.3 |
| Same models on Snapdragon 8 Elite phones               | 95      | 3      | 60%   | 4           | 42.8 |
| Voice-clone (deepfake) detector on NPU (WavLM, AI Hub) | 60      | 3      | 50%   | 3           | 30   |

Timeline

Now (v1.0) 3 sentinels, 5 NPU-ready models, fusion, dashboard, 27 tests

+1 month on-device benchmark on X Elite, real-call pilot, Marathi

+3 months voice-clone detector, family guardian, installer (MSIX)

+6 months Android build on Snapdragon 8 Elite, bank / OEM pilots

| Risks and how Sajag handles them |                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------|
| Risk                             | Mitigation                                                                                             |
| False alarms erode trust         | awareness damping, classifier-weighted evidence, hard-negative corpus; 0.5% synthetic false-alarm rate |
| ASR errors on noisy calls        | screen + system evidence can confirm or replace audio; forced language prompt                          |
| Scammers change scripts          | tactic-level (not keyword-level) model; cue library and classifier update offline                      |
| Privacy perception               | open source, no network at runtime, nothing stored, verifiable by anyone                               |

Business impact vs effort

Real-call validation

Phone port

More languages

Voice-clone detector

Family guardian

OEM preload pilot

MSIX installer

Top row: higher impact | Bottom row: lower impact | Left: low effort | Right: high effort



# THANK YOU

धन्यवाद

Sajag AI turns every Snapdragon AI PC into a private, always-alert guardian against the scams costing Indians thousands of crores a year.

#### Code

[github.com/tanmaysahare/sajag-ai](https://github.com/tanmaysahare/sajag-ai)

#### Try it

`python -m sajag demo`

#### Author

Tanmay Sahare, IIT Kanpur

#### If you are being scammed

Call 1930 | [cybercrime.gov.in](https://cybercrime.gov.in)



# Appendix A: References (1-52 of 156)

Full annotated list in the project description and docs/REFERENCES.md

[1] Snapdragon AI Lab Build & Present Challenge 2026, Unstop listing.  
<https://unstop.com/competitions/crp-snapdragon-ai-lab-build-present-challenge-qualcomm-1748893>

[2] Snapdragon AI Lab, Qualcomm. <https://www.qualcomm.com/snapdragon/ai-lab>

[3] Snapdragon AI Lab Build & Present Challenge 2026, aicompetition.dev summary.  
<https://www.aicompetition.dev/en/competitions/snapdragon-ai-lab-build-present-2026>

[4] Snapdragon X Elite product page, Qualcomm.  
<https://www.qualcomm.com/laptops/products/snapdragon-x-elite>

[5] On-device AI laptops powered by Snapdragon, Qualcomm.  
<https://www.qualcomm.com/snapdragon/laptops-and-tablets/consumer/ai>

[6] Snapdragon X Elite vs X Plus, HP Tech Takes.  
<https://www.hp.com/us-en/tech-takes/components/comparison/snapdragon-x-elite-vs-x-plus-difference.html>

[7] Snapdragon X Elite powers HP OmniBook X, HP Tech Takes. <https://www.hp.com/gb-en/tech-takes/laptops/review/snapdragon-x-elite-omnibook-x-review.html>

[8] HP OmniBook Ultra 14 (Snapdragon X2) review, Tom's Hardware.  
<https://www.tomshardware.com/laptops/hp-omnibook-ultra-14-snapdragon-x2-elite-2026-review>

[9] HP OmniBook Ultra 14 gets Snapdragon X2 options (85 TOPS variant), Tom's Hardware.  
<https://www.tomshardware.com/laptops/hps-new-omnibook-ultra-14-gets-panther-lake-and-snapdragon-x2-options-inside-exclusive-variant-of-qualcomm-chip-has-85-tops>

[10] HP OmniBook Ultra 14 (Snapdragon X2) review, Thurrott.  
<https://www.thurrott.com/mobile/copilot-pc/340107/hp-omnibook-ultra-14-snapdragon-x2-review>

[11] All Snapdragon X2 Elite laptops announced at CES, XDA.  
<https://www.xda-developers.com/snapdragon-x2-elite-laptops-ces/>

[12] Snapdragon X2 Elite laptops 2026, The Gadgeteer.  
<https://the-gadgeteer.com/2026/05/13/snapdragon-x2-elite-laptops-2026/>

[13] GenieX on Hexagon NPU for local agentic tasks on Snapdragon X/X2 Elite, Qualcomm Developer Blog. <https://www.qualcomm.com/developer/blog/2026/09/clairvoyance-genie-x-hexagon-snapdragon>

[14] Smart routing with Snapdragon X Series NPUs, Qualcomm Developer Blog.  
<https://www.qualcomm.com/developer/blog/2025/08/smart-routing-snapdragon-x-series-npu-developer-perspective-teknikos>

[15] Qualcomm AI Hub will bring Stable Diffusion, Whisper and more, XDA. <https://www.xda-developers.com/qualcomm-ai-hub-snapdragon-announced/>

[16] Qualcomm launches AI Hub with optimized on-device models, Hackster.  
<https://www.hackster.io/news/qualcomm-launches-its-ai-hub-offering-optimized-on-device-ai-models-with-four-times-the-performance-3dddfcb8a84f>

[17] Copilot+ PCs developer guide, Microsoft Learn.  
<https://learn.microsoft.com/en-us/windows/ai/npu-devices/>

[18] Everything about Copilot+ PCs with Snapdragon X, ElevenForum.  
<https://www.elevenforum.com/t/everything-you-need-to-know-about-copilot-pcs-by-snapdragon-x-series-processors.25813/>

[19] Snapdragon X Copilot+ PC app compatibility scorecard, Thurrott.  
<https://www.thurrott.com/mobile/copilot-pc/304599/snapdragon-x-copilot-pc-my-windows-11-on-arm-app-compatibility-scorecard>

[20] Arm64 Copilot+ PC or x86, Windows Central.  
<https://www.windowscentral.com/hardware/laptops/arm64-copilot-plus-pc-or-intel-amd-nvidia>

[21] Running small language models on the NPU with Copilot+ PCs.  
<https://pkbullock.com/blog/2025/running-slm-models-using-npu-with-copilot-pc>

[22] Qualcomm AI Hub Models repository, GitHub. <https://github.com/qualcomm/ai-hub-models>

[23] qai-hub-models, PyPI. <https://pypi.org/project/qai-hub-models/>

[24] Qualcomm AI Hub documentation overview. <https://app.aihub.qualcomm.com/docs/hub/>

[25] Qualcomm AI Hub: compiling models.  
[https://workbench.aihub.qualcomm.com/docs/hub/compile\\_examples.html](https://workbench.aihub.qualcomm.com/docs/hub/compile_examples.html)

[26] Qualcomm AI Hub: working with frameworks.  
<https://app.aihub.qualcomm.com/docs/hub/frameworks.html>

[27] Qualcomm AI Hub: API documentation.  
<https://workbench.aihub.qualcomm.com/docs/hub/api.html>

[28] Qualcomm AI Hub: FAQ. <https://app.aihub.qualcomm.com/docs/hub/faq.html>

[29] Qualcomm AI Hub: release notes.  
[https://app.aihub.qualcomm.com/docs/hub/release\\_notes.html](https://app.aihub.qualcomm.com/docs/hub/release_notes.html)

[30] Whisper-Small model card, Qualcomm AI Hub.  
[https://aihub.qualcomm.com/models/whisper\\_small](https://aihub.qualcomm.com/models/whisper_small)

[31] Whisper-Large-v3-Turbo model card, Qualcomm AI Hub.  
[https://aihub.qualcomm.com/models/whisper\\_large\\_v3\\_turbo](https://aihub.qualcomm.com/models/whisper_large_v3_turbo)

[32] Silero-VAD model card, Qualcomm AI Hub. [https://aihub.qualcomm.com/models/silero\\_vad](https://aihub.qualcomm.com/models/silero_vad)

[33] EasyOCR model card, Qualcomm AI Hub. <https://aihub.qualcomm.com/models/easyocr>

[34] Llama-v3.2-3B-Instruct model card, Qualcomm AI Hub.  
[https://aihub.qualcomm.com/compute/models/llama\\_v3\\_2\\_3b\\_instruct](https://aihub.qualcomm.com/compute/models/llama_v3_2_3b_instruct)

[35] Nomic-Embed-Text model card, Qualcomm AI Hub.  
[https://aihub.qualcomm.com/models/nomic\\_embed\\_text](https://aihub.qualcomm.com/models/nomic_embed_text)

[36] WavLM-Base-Plus model card, Qualcomm AI Hub.  
[https://aihub.qualcomm.com/models/huggingface\\_wavlm\\_base\\_plus](https://aihub.qualcomm.com/models/huggingface_wavlm_base_plus)

[37] Qwen3-4B-Instruct-2507 model card, Qualcomm AI Hub.  
[https://aihub.qualcomm.com/models/qwen3\\_4b\\_instruct\\_2507](https://aihub.qualcomm.com/models/qwen3_4b_instruct_2507)

[38] Prepare ONNX models, Qualcomm AI Developer Workflow.  
<https://docs.qualcomm.com/doc/80-80022-15B/topic/onnx-prepare-model.html>

[39] Running precompiled QNN ONNX / context binaries in ONNX Runtime, Qualcomm support forum. <https://mysupport.qualcomm.com/supportforums/s/question/0D5dK0000Bvt1WSAR/how-can-i-run-inference-in-onnx-runtime-qnn-using-a-precompiledqnnonnx-model-or-a-qnncontextbinary-generated-by-the-qaihub-api>

[40] QNN Execution Provider, ONNX Runtime documentation.  
<https://onnxruntime.ai/docs/providers/QNN-ExecutionProvider.html>

[41] onnxruntime-qnn repository, GitHub. <https://github.com/onnxruntime/onnxruntime-qnn>

[42] QNN Execution Provider docs (onnxruntime-qnn).  
[https://github.com/onnxruntime/onnxruntime-qnn/blob/main/docs/execution\\_providers/QNN-ExecutionProvider.md](https://github.com/onnxruntime/onnxruntime-qnn/blob/main/docs/execution_providers/QNN-ExecutionProvider.md)

[43] ONNX Runtime QNN EP issue on Snapdragon X Elite, GitHub #22617.  
<https://github.com/microsoft/onnxruntime/issues/22617>

[44] QNN Execution Provider guide, Radxa. [https://docs.radxa.com/en/fogwise/airbox-q900/ai-dev/qnn\\_onnxrt\\_execution\\_provider](https://docs.radxa.com/en/fogwise/airbox-q900/ai-dev/qnn_onnxrt_execution_provider)

[45] YOLO export to Qualcomm QNN (Snapdragon), Ultralytics.  
<https://docs.ultralytics.com/integrations/qnn>

[46] Running YOLOv8 on Snapdragon X Elite NPU at 65 FPS, Medium.  
<https://medium.com/@carrycoidude/unleashing-the-beast-running-yolov8-on-snapdragon-x-elite-npu-at-65-fps-296084253c2c>

[47] LLM on Genie tutorial, Qualcomm AI Hub Apps.  
[https://github.com/qualcomm/ai-hub-apps/tree/main/tutorials/llm\\_on\\_genie](https://github.com/qualcomm/ai-hub-apps/tree/main/tutorials/llm_on_genie)

[48] How to use Genie, Qualcomm AI Runtime (QAI RT) SDK. [https://docs.qualcomm.com/doc/80-63442-10/topic/how\\_to\\_use\\_genie.html](https://docs.qualcomm.com/doc/80-63442-10/topic/how_to_use_genie.html)

[49] Qualcomm Genie on Windows on Snapdragon.  
<https://docs.qualcomm.com/doc/80-62010-1/topic/genie.html>

[50] GenieX: run LLMs and VLMs locally on Snapdragon, GitHub.  
<https://github.com/qualcomm/geniex>

[51] Offline note taker on Snapdragon X Elite with Whisper and Qwen, Medium.  
<https://shivaylamba.medium.com/offline-note-taker-building-private-meeting-notes-on-snapdragon-x-elite-with-whisper-qwen-and-be1090adefff>

[52] Whispp and Qualcomm voice reconstruction AI on Snapdragon X Elite.  
<https://whispp.com/whitepapers/whispp-qualcomm-voice-reconstruction-ai-snapdragon-x-elite/>



# Appendix B: References (53-104 of 156)

Full annotated list in the project description and docs/REFERENCES.md

[53] Whisper-large-v3-turbo QNN build for Snapdragon X Elite, Hugging Face. <https://huggingface.co/FluidInference/whisper-large-v3-turbo-qnn/blob/main/snapdragon-x-elite/README.md>

[54] Radford et al., Robust Speech Recognition via Large-Scale Weak Supervision (Whisper), arXiv 2212.04356. <https://arxiv.org/abs/2212.04356>

[55] Whisper paper PDF, OpenAI. <https://cdn.openai.com/papers/whisper.pdf>

[56] OpenAI Whisper model card, Hugging Face. <https://huggingface.co/openai/whisper-large>

[57] whisper.cpp, GitHub. <https://github.com/ggml-org/whisper.cpp>

[58] WhisperLive, Collabora, GitHub. <https://github.com/collabora/WhisperLive>

[59] Silero VAD, GitHub. <https://github.com/snakers4/silero-vad>

[60] Silero VAD wiki. <https://github.com/snakers4/silero-vad/wiki/>

[61] Silero models, GitHub. <https://github.com/snakers4/silero-models>

[62] Du et al., PP-OCR: A Practical Ultra Lightweight OCR System, arXiv 2009.09941. <https://arxiv.org/abs/2009.09941>

[63] Li et al., PP-OCRv3: More Attempts for the Improvement of Ultra Lightweight OCR, arXiv 2206.03001. <https://arxiv.org/abs/2206.03001>

[64] PaddleOCR 3.0 Technical Report. <https://www.alphaxiv.org/abs/2507.05595>

[65] InsightFace ArcFace on Qualcomm NPU example, GitHub. <https://github.com/Soorya2201/qualcomm-insightface-npu>

[66] Llama-3.2-3B Genie-compatible QNN binaries, Hugging Face. <https://huggingface.co/Volk76/Llama-3.2-3B-Genie-Compatible-QNN-Binaries>

[67] 'It Warned Me Just at the Right Moment': LLM-based real-time detection of phone scams, arXiv 2502.03964. <https://arxiv.org/abs/2502.03964>

[68] Combating Phone Scams with LLM-based Detection: Where Do We Stand?, arXiv 2409.11643. <https://arxiv.org/abs/2409.11643>

[69] AI-in-the-Loop: Privacy Preserving Real-Time Scam Detection and Conversational Scambaiting, arXiv 2509.05362. <https://arxiv.org/abs/2509.05362>

[70] Before the Warning Comes Too Late: Incremental Phone-Scam Detection, arXiv 2609.20223. <https://arxiv.org/html/2609.20223>

[71] ScamAgents: How AI Agents Can Simulate Human-Level Scam Calls, arXiv 2508.06457. <https://arxiv.org/html/2508.06457>

[72] Advanced Real-Time Fraud Detection Using RAG-Based LLMs, arXiv 2501.15290. <https://arxiv.org/html/2501.15290v1>

[73] MHA Lok Sabha reply on cyber fraud (Dec 2025). <https://www.mha.gov.in/MHA1/Par2017/pdfs/par2025-pdfs/LS02122025/432.pdf>

[74] MHA Rajya Sabha reply on cyber crime (Feb 2026). <https://www.mha.gov.in/MHA1/Par2017/pdfs/par2026-pdfs/RS04022026/553.pdf>

[75] MHA Lok Sabha reply on cyber crime (Jul 2025). <https://xn--i1b5zbzybhfo5c8b4bxh.xn--11b7cb3a6a.xn--h2brj9c/MHA1/Par2017/pdfs/par2025-pdfs/LS22072025/344.pdf>

[76] MHA Rajya Sabha reply on digital arrest (Mar 2025). <https://xn--i1b5zbzybhfo5c8b4bxh.xn--11b7cb3a6a.xn--h2brj9c/MHA1/Par2017/pdfs/par2025-pdfs/RS12032025/1513.pdf>

[77] Indians lost Rs 22,845 crore to cyber fraud in 2024, a 206% rise, Scroll. <https://scroll.in/latest/1084796/indians-lost-rs-22845-crore-to-cyber-fraud-in-2024-a-206-rise-from-previous-year-centre>

[78] After 465% spike in 2024, MHA data shows digital arrest scams on a decline, ThePrint. <https://theprint.in/india/governance/after-465-spike-in-2024-mha-data-shows-digital-arrest-scams-are-on-a-decline-in-india/2857002/>

[79] India lost Rs 11,333 crore to cyber fraud in 2024 (first 9 months), The Wire. <https://m.thewire.in/article/tech/india-lost-rs-11333-crore-to-cyber-fraud-in-2024>

[80] Scammers steal Rs 2,140 crore in 10 months using fake digital arrests, India TV. <https://www.indiatvnews.com/technology/news/scammers-steal-rs-2-140-crore-in-10-months-using-fake-digital-arrests-mha-report-2024-11-01-959766>

[81] CyberPeace analysis of India's cybercrime surge from MHA data. <https://cyberpeace.org/resources/blogs/cyberpeace-analysis-indias-cybercrime-surge-signals-a-growing-digital-security-challenge-an-assessment-based-on-rajya-sabha-proceedings-and-mha-data>

[82] How much Indians lost to cyber frauds Jan-Apr 2024, Business Standard. [https://www.business-standard.com/india-news/here-is-how-much-indians-lost-to-cyber-frauds-between-jan-and-apr-of-2024-124052700151\\_1.html](https://www.business-standard.com/india-news/here-is-how-much-indians-lost-to-cyber-frauds-between-jan-and-apr-of-2024-124052700151_1.html)

[83] Cyber fraud losses could amount to 0.7% of GDP (I4C), Manorama Yearbook. <https://www.manoramayearbook.in/current-affairs/india/2024/10/26/i4c-cyber-fraud-losses.html>

[84] Digital arrest: Rs 120 crore lost Jan-Apr 2024, MHA advisory, Organiser. <https://organiser.org/2024/10/28/262662/bharat/digital-arrest-rs-120-crore-lost-in-cybercrimes-between-january-april-2024-mha-calls-for-citizen-vigilance/>

[85] Cyber frauds cost India Rs 22,845 crore in 2024, Digital Terminal. <https://digitalterminal.in/trending/cyber-frauds-cost-india-22845-crore-in-2024-as-digital-threats-escalate>

[86] Cyber fraud costs Indians Rs 22,845 crore in 2024, Angel One. <https://www.angelone.in/news/market-updates/cyber-fraud-costs-indians-22-845-crore-in-2024-government>

[87] Cyber fraud cases surge in 2024, Outlook Money. <https://www.outlookmoney.com/news/cyber-fraud-cases-surge-in-2024-citizens-lost-rs-22845-crore-says-govt>

[88] Big surge in cyber fraud complaints, Industrial Economist. <https://industrialeconomist.com/cyber-fraud-complaints-surge-in-2024-losses-cross-%E2%82%B922800-cr/>

[89] Complaints rose fivefold since 2021, IndiaSpend. <https://www.indiaspend.com/data-viz/dataviz-how-indias-cyber-crime-incidence-is-rising-972933>

[90] Digital arrest scam, PIB press release. <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=2082761&reg=48&lang=2>

[91] Cyber fraud and digital harassment, PIB. <https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2080186>

[92] Cyber awareness, PIB. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2241336&reg=3&lang=1>

[93] PIB press note on cyber crime. <https://www.pib.gov.in/PressNoteDetails.aspx?Noteld=155384&ModuleId=3&reg=48&lang=2>

[94] Recent cybercrime trends in financial frauds, NCERT CIET workshop (Aug 2024). [https://ciet.ncert.gov.in/storage/app/public/files/17/Workshop\\_and\\_training\\_files/fsch/12aug.pdf](https://ciet.ncert.gov.in/storage/app/public/files/17/Workshop_and_training_files/fsch/12aug.pdf)

[95] Citizen Financial Cyber Fraud Reporting and Management System, Vikaspedia. <https://egovernance.vikaspedia.in/viewcontent/e-governance/citizen-services/citizen-financial-cyber-fraud-reporting-and-management-system?lgn=en>

[96] Digital arrest scam, Wikipedia. [https://en.wikipedia.org/wiki/Digital\\_arrest\\_scam](https://en.wikipedia.org/wiki/Digital_arrest_scam)

[97] What is the digital arrest scam, Forbes India. <https://www.forbesindia.com/article/explainers/what-is-the-digital-arrest-scam-and-how-to-be-aware/2988963/1>

[98] Digital arrest scams and the limits of domestic enforcement, ORF. <https://www.orfonline.org/expert-speak/digital-arrest-scams-and-the-limits-of-domestic-enforcement>

[99] India's digital arrest scams, Lowy Institute. <https://www.loyyinstitute.org/the-interpreter/india-s-digital-arrest-scams>

[100] Digital arrest scams explained, PMF IAS. <https://www.pmfias.com/digital-arrest/>

[101] Mann Ki Baat: PM warns of digital arrest, Telangana Today. <https://telanganatoday.com/mann-ki-baat-pm-modi-warns-of-digital-arrest-urges-people-to-stay-vigilant>

[102] Ruko, socho, action lo: PM's mantra against digital arrest, Newstrack. <https://newstrack.com/india/wait-think-and-take-action-pm-gave-mantra-to-avoid-digital-arrest-475671>

[103] 50,000 appeals pending before grievance panel, victims navigate maze of portals, ThePrint. <https://theprint.in/india/50000-appeals-pending-before-govts-social-media-grievance-panel-victims-navigate-maze-of-portals/3057672/>

[104] 1 lakh seniors, 4.6 lakh women lost Rs 7,769 crore to cyber fraud, NewKerala. <https://www.newkerala.com/news/a/over-lakh-senior-citizens-463-lakh-women-lodged-614.htm>





# Appendix C: References (105-156 of 156)

Full annotated list in the project description and docs/REFERENCES.md

[105] Hyderabad senior citizens lose Rs 102 crore to cyber fraud in 19 months, NewsMeter. <https://newsmeter.in/top-stories/hyderabad-senior-citizens-lose-rs-102-crore-to-cyber-fraud-in-19-months-retiree-duped-of-rs-96-lakh-773304>

[106] Cyber fraud steals Rs 1,250 crore in Delhi in one year, Asianet. <https://newsable.asianetnews.com/india/cyber-fraud-losses-surge-in-delhi-digital-arrest-scams-drain-life-savings-articleshow-cp974wg>

[107] Nagpur cyber fraud 2025: 60% of victims senior citizens, The420. <https://the420.in/nagpur-cyber-fraud-2025-senior-citizens-digital-arrest-scams/>

[108] Indore: over Rs 100 crore lost to cyber fraud in 2025, The420. <https://the420.in/cyber-fraud-indore-2025-100-crore-loss-digital-arrest-seniors-targeted/>

[109] Elderly woman loses Rs 2.5 crore in digital arrest fraud, Tribune. <https://www.tribuneindia.com/news/chandigarh/sector-10-elderly-woman-loses-2-5-crore-in-digital-arrest-fraud>

[110] Digital arrest scam targeting India's elderly parents, Gracias Living. <https://graciasliving.com/digital-arrest-scam-elderly-parents-india/>

[111] Microsoft Global Tech Support Scam Research: India (2021). <https://news.microsoft.com/wp-content/uploads/prod/sites/45/2021/07/Microsoft-India-Tech-Support-Fraud-Survey-2021.pdf>

[112] Seven out of 10 consumers in India encountered tech support scams, Microsoft News. <https://news.microsoft.com/en-in/seven-out-of-10-consumers-in-india-encountered-tech-support-scams-in-the-past-year-microsoft-survey/>

[113] Indians more vulnerable to tech support scams, Business Standard. [https://www.business-standard.com/article/technology/indians-more-vulnerable-to-tech-support-scams-says-microsoft-report-121072201212\\_1.html](https://www.business-standard.com/article/technology/indians-more-vulnerable-to-tech-support-scams-says-microsoft-report-121072201212_1.html)

[114] India sees the highest number of tech support scams, CXOToday. <https://cxotoday.com/daily-news/india-sees-the-highest-number-of-tech-support-scams-reveals-microsoft/>

[115] Almost half of Indians experience AI-enabled fake voice scam: McAfee, Business Standard. [https://www.business-standard.com/india-news/almost-half-of-indians-experience-ai-enabled-fake-voice-scam-mcafee-survey-123050100949\\_1.html](https://www.business-standard.com/india-news/almost-half-of-indians-experience-ai-enabled-fake-voice-scam-mcafee-survey-123050100949_1.html)

[116] 83% Indians fell prey to AI voice scams: McAfee report, BOOM. <https://www.boomlive.in/explainers/ai-voice-scam-voice-cloning-mcafee-report-21851>

[117] Artificial imposters: cybercriminals turn to AI voice cloning, McAfee. <https://www.mcafee.com/blogs/privacy-identity-protection/artificial-imposters-cybercriminals-turn-to-ai-voice-cloning-for-a-new-breed-of-scam/>

[118] Has AI become a powerful tool for scamming?, IndiaAI. <https://indiaai.gov.in/article/has-ai-really-become-a-powerful-tool-for-scamming>

[119] RBI warns of fraud that leverages AnyDesk, BankInfoSecurity. <https://www.bankinfosecurity.asia/rbi-warns-fraud-that-leverages-anydesk-app-a-12035>

[120] RBI warns digital and UPI users against AnyDesk, Business Standard. [https://www.business-standard.com/article/finance/digital-banking-upi-users-beware-rbi-warns-against-anydesk-all-you-need-to-know-119021800425\\_1.html](https://www.business-standard.com/article/finance/digital-banking-upi-users-beware-rbi-warns-against-anydesk-all-you-need-to-know-119021800425_1.html)

[121] After RBI, NPCI warns about AnyDesk app fraud, Zee Business. <https://www.zeebiz.com/india/news-after-rbi-upi-operator-ncpi-warns-about-anydesk-app-fraud-how-your-money-is-robbed-and-how-to-save-it-86540>

[122] Phishing, UPI scams and remote access fraud on the rise, The420. <https://the420.in/online-banking-fraud-phishing-upi-remote-access/>

[123] RBI Annual Report 2024-25 highlights fraud surge, MediaNama. <https://www.medianama.com/2025/06/223-rbi-annual-report-2024-25-central-bank-agenda/>

[124] Bank frauds rise 166% in FY24 to over 36,000 cases, Business Standard. [https://www.business-standard.com/industry/banking/bank-fraud-rise-166-in-fy24-to-over-36-000-cases-shows-rbi-annual-report-124053001237\\_1.html](https://www.business-standard.com/industry/banking/bank-fraud-rise-166-in-fy24-to-over-36-000-cases-shows-rbi-annual-report-124053001237_1.html)

[125] Banking frauds rise in H1FY25, amount jumps eightfold, Business Standard. [https://www.business-standard.com/industry/banking/banking-fraud-rise-in-h1fy25-amount-involved-jumps-8-time-rbi-report-124122600769\\_1.html](https://www.business-standard.com/industry/banking/banking-fraud-rise-in-h1fy25-amount-involved-jumps-8-time-rbi-report-124122600769_1.html)

[126] RBI: online payment fraud surges 400% in a year, PYMNTS. <https://www.pymnts.com/news/security-and-risk/2024/rbi-online-payment-fraud-in-india-surges-400percent-in-a-year>

[127] Stand up to scareware with scareware blocker, Microsoft Edge blog (Jan 2025). <https://blogs.windows.com/msedgedev/2025/01/27/stand-up-to-scareware-with-scareware-blocker/>

[128] Expanded scareware blocker availability and real-time protection, Microsoft Edge blog (Oct 2025). <https://blogs.windows.com/msedgedev/2025/10/31/protecting-more-edge-users-with-expanded-scareware-blocker-availability-and-real-time-protection/>

[129] Prevent online scams with the scareware blocker in Microsoft Edge, Microsoft Support. <https://support.microsoft.com/en-us/topic/prevent-online-scams-with-the-scareware-blocker-in-microsoft-edge-b02c7895-f9b7-4d9f-8e12-3668f00915be>

[130] Microsoft tests Edge scareware blocker, BleepingComputer. <https://www.bleepingcomputer.com/news/microsoft/microsoft-tests-edge-scareware-blocker-to-block-tech-support-scams/>

[131] Microsoft Edge gets scareware sensor, BleepingComputer. <https://www.bleepingcomputer.com/news/microsoft/microsoft-edge-gets-scareware-sensor-for-faster-scam-detection/>

[132] Google rolls out on-device AI protections to detect scams in Chrome and Android, The Hacker News. <https://thehackernews.com/2025/05/google-rolls-out-on-device-ai.html>

[133] Google deploys on-device AI to thwart scams on Chrome and Android, Infosecurity. <https://www.infosecurity-magazine.com/news/google-ai-gemini-nano-scams-chrome/>

[134] Google's Safety Charter for India's AI-led transformation, Google India blog. <https://blog.google/intl/en-in/company-news/googles-safety-charter-for-indias-ai-led-transformation/>

[135] Protecting vulnerable audiences is at the heart of AI safety efforts, Google India blog. <https://blog.google/intl/en-in/company-news/protecting-vulnerable-audiences-is-at-the-heart-of-ai-safety-efforts/>

[136] Google steps up fight against online fraud with India Safety Charter, Tribune. <https://www.tribuneindia.com/news/india/google-steps-up-fight-against-online-ai-fraud-deepfakes-with-india-safety-charter/>

[137] Google steps up AI scam protection in India, but gaps remain, Yahoo Tech. <https://tech.yahoo.com/ai/gemini/articles/google-steps-ai-scam-protection-222852933.html>

[138] Microsoft SmartScreen, Wikipedia. [https://en.wikipedia.org/wiki/Microsoft\\_SmartScreen](https://en.wikipedia.org/wiki/Microsoft_SmartScreen)

[139] Digital Personal Data Protection Rules 2025, PIB note. <https://www.pib.gov.in/PressNoteDetails.aspx?NetId=156054&ModuleId=3&reg=3&lang=2>

[140] DPDP Rules 2025 notified, PIB document. <https://static.pib.gov.in/WriteReadData/specifdoc/docs/documents/2025/nov/doc20251117695301.pdf>

[141] Digital Personal Data Protection Rules, 2025, Wikipedia. [https://en.wikipedia.org/wiki/Digital\\_Personal\\_Data\\_Protection\\_Rules,\\_2025](https://en.wikipedia.org/wiki/Digital_Personal_Data_Protection_Rules,_2025)

[142] DPDP Rules 2025 notified by MeitY, EY. [https://www.ey.com/en\\_in/insights/cybersecurity/transforming-data-privacy-digital-personal-data-protection-rules-2025](https://www.ey.com/en_in/insights/cybersecurity/transforming-data-privacy-digital-personal-data-protection-rules-2025)

[143] DPDP Rules 2025 key highlights, SCC Online. <https://www.scconline.com/blog/post/2025/12/26/digital-personal-data-protection-rules-2025-key-highlights/amp/>

[144] IDC: India's PC market grows 5.7% YoY in 1H25. <https://my.idc.com/getdoc.jsp?containerId=prAP53752625>

[145] India PC shipments hit record 15.9 million in 2025, ChannelLife. <https://channellife.in/story/india-pc-shipments-hit-record-15-9-million-in-2025>

[146] Indian PC market grows 10.2% YoY in 2025 (IDC), FoneArena. <https://www.fonearena.com/blog/476970/indian-pc-market-2025-idc.html>

[147] HP leads India's PC market in Q3 2025 with 26.6% share, Digital Terminal. <https://digitalterminal.in/trending/hp-leads-indias-pc-market-in-q3-2025-with-266-market-share>

[148] HP leads Indian PC market with 29.1% share, ETV Bharat. <https://www.etvbharat.com/en/!technology/hp-leads-indian-pc-market-with-29-pc-share-lenovo-grows-to-claim-second-spot-idc-enn25052701571>

[149] SnapShield (challenge entry, network security), GitHub. <https://github.com/CODEwANI10/Qualcomm-AI-Build-Challenge-Snapshield>

[150] SnapStudy-AI (challenge entry, study assistant), GitHub. <https://github.com/nikhilmohite120625/SnapStudy-AI>

[151] ShikshaMate (challenge entry, education), GitHub. <https://github.com/infant88/ShikshaMate>

[152] OmniCare-AI (challenge entry, healthcare), GitHub. <https://github.com/Advik-harsha/OmniCare-AI>

[153] SnapCareer-AI (challenge entry, careers), GitHub. <https://github.com/Mukeshpadigela17/SnapCareer-AI-project>

[154] snapdragoon (challenge entry, accessibility), GitHub. <https://github.com/challalokesh08/snapdragoon>

[155] Snapdragon AI Lab Challenge 2026 overview video, YouTube. [https://www.youtube.com/watch?v=ZTT\\_be\\_nkv8](https://www.youtube.com/watch?v=ZTT_be_nkv8)

[156] Qualcomm India opens Snapdragon Multiverse Hackathon registrations, The Wire. <https://m.thewire.in/article/ptiprnews/qualcomm-india-opens-registrations-for-snapdragon-multiverse-hackathon-to-build-multi-device-ai-solutions>